

Paper 2

Title

Data-Driven Prediction of Dissolved Oxygen for Fish Kill Identification Using CNN-GRU Deep Learning

Abstract

Dissolved Oxygen (DO) is one of the most important indicators of aquatic ecosystem health and plays a critical role in preventing fish kill events. Predicting future DO levels enables early intervention before environmental conditions become harmful to aquatic organisms. This study proposes a hybrid Convolutional Neural Network–Gated Recurrent Unit (CNN-GRU) model for forecasting dissolved oxygen concentrations in the Redbank Weir of the Murrumbidgee River, Australia. Multiple environmental datasets, including hydrological measurements, meteorological variables, thermal stratification, satellite-derived chlorophyll information, and climatic conditions, were integrated to build the predictive model. Data preprocessing techniques, including missing value imputation and normalization, were applied before model training. The proposed CNN-GRU model outperformed several conventional deep learning models in both hourly and daily DO prediction. Furthermore, Explainable Artificial Intelligence (XAI) techniques were used to identify the environmental variables contributing most significantly to dissolved oxygen fluctuations. The study demonstrates that combining deep learning with explainability provides valuable insights for water quality management and helps identify environmental conditions leading to fish kill events.

Introduction

Dissolved oxygen is one of the most critical parameters governing aquatic ecosystem health because it directly influences fish respiration, biological activity, and water quality. Low dissolved oxygen concentrations often result in hypoxic conditions, leading to large-scale fish mortality events that negatively affect both ecological sustainability and the aquaculture industry.

Predicting dissolved oxygen is challenging because it is influenced by multiple interacting environmental variables, including water temperature, evaporation, rainfall, chlorophyll concentration, thermal stratification, humidity, solar radiation, and water levels. Traditional hydrodynamic and water-quality simulation models require extensive domain expertise and large numbers of physical parameters, making them difficult to implement in practical situations.

With recent advances in artificial intelligence, deep learning has become a promising alternative for modeling complex nonlinear relationships among environmental variables. In this study, the authors developed a hybrid CNN-GRU architecture capable of learning both spatial and temporal

characteristics of environmental data. The model utilized hydrological, meteorological, satellite, and climate datasets collected over several years to forecast dissolved oxygen concentrations at both hourly and daily intervals.

To improve model interpretability, Explainable AI methods were employed to analyze feature importance, revealing that surface water temperature, evaporation, thermal stratification, and chlorophyll concentration were among the major contributors influencing dissolved oxygen dynamics. The study demonstrates that accurate dissolved oxygen prediction can serve as an effective early indicator of potential fish kill events and improve aquatic ecosystem management.